Statistika, Institut Teknologi
Sepuluh Nopember Surabaya

Addressing Outliers in Customer Segmentation: Comparative Effectiveness of K-Prototype Clustering and Mix Type DBSCAN

Study Case : Supplement Retail

Dosen Pengampu

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Overview

The main part of this presentation

01 Introduction

02 Problem

03 Algorithm

04 Methods

05 Analysis

06 Results and Recomendation

Introduction

Background Study

Healthy awareness

Consumers are becoming more aware of health and fitness

Growth of the Supplement Industry

Grand View Research Report (2020): The global supplement market is projected to reach USD 278.02 billion by 2024 with a CAGR of 8.2% from 2019 to 2024.

Trends in Indonesia

Nielsen Study (2019): Indonesian consumers are increasingly health-conscious and prefer products that support their health.

Challenges in the Supplement Industry

Intense competition among supplement stores requires product and service innovation.

Importance of Accurate Market Segmentation

Proper market segmentation allows for a deeper understanding of consumer needs and preferences.

Analytical Approach to Understanding Customers

Supplement stores need to adopt more sophisticated analytical approaches

- K-PROTOYPE CLUSTERING
- MIX TYPE DBSCAN (Gower Distance)

Type of Data : Categorical and Numerical
There outliers in Numerical Variables

Algorithm

Outliers

- Outliers are data points or observations that have values significantly different from other observations.
- Causes of outliers include measurement errors, data entry errors, rare events, and other factors

Detection Method

- Graphical methods: Boxplot
- Descriptive statistics: Z-score and Interquartile Range (IQR)
- Statistical tests: Grubb's test, Dixon test

Handling Method

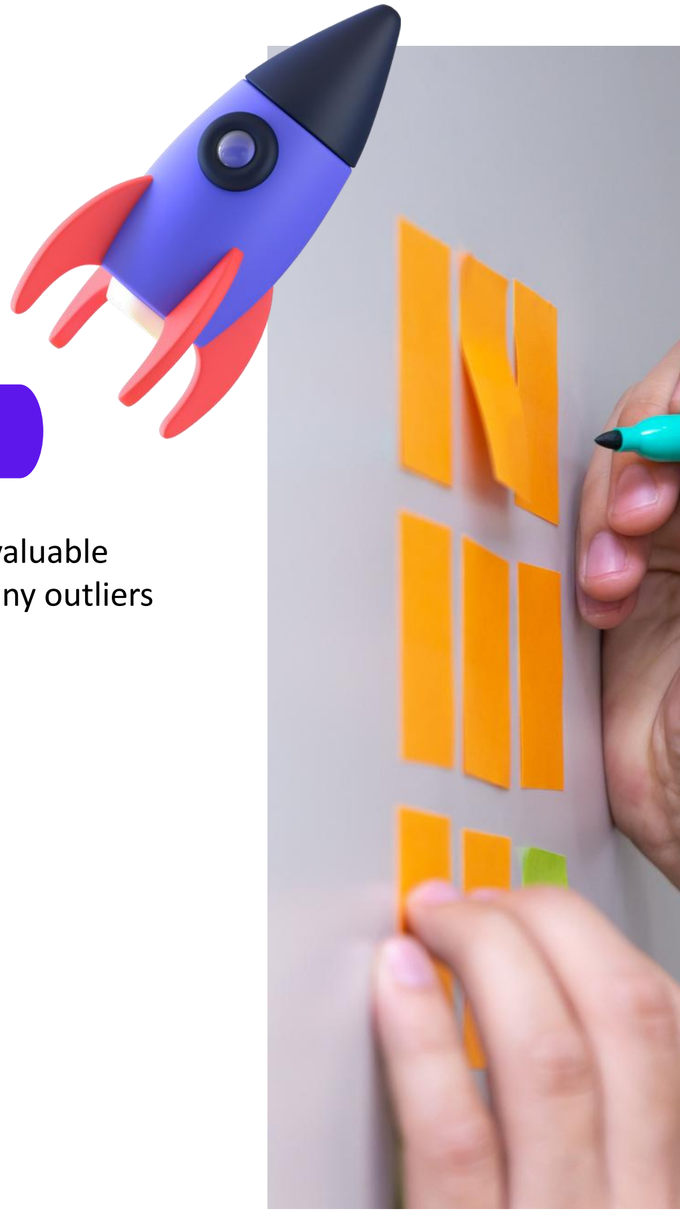
- Deleting outliers
- Data transformation
- Imputation
- Winsorizing (Dash et al., 2023)

Winsorizing replaces outliers with values at the 10th percentile for lower outliers and the 90th percentile for upper outliers.

Percentile thresholds can be adjusted based on data needs (Sihombing et al., 2023)

Drawbacks of Deleting

Removing outliers can lead to loss of valuable information, especially if there are many outliers in the dataset



K-prototype

- The K-Prototype algorithm is a cluster analysis where the data is a mixture of numerical and categorical data
- This algorithm is a mixture of K-means and K-Modes, which respectively are cluster analysis used on numerical and categorical data.

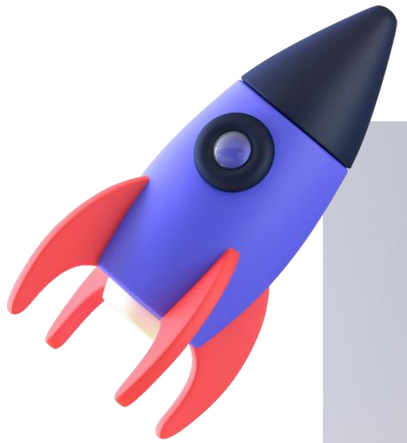
NUMERICAL VARIABLE

$$d_r(X_j, Z_i) = \sqrt{\sum_{i=1}^{n} (x_{jk} - z_{ik})^2}$$

CATEGORICAL VARIABLE

$$d_c(X_j, Z_i) = \gamma_i \sum_{p=k+1}^{m} \delta(x_{jk}, z_{ik})$$

$$= \sqrt{\sum_{i=1}^{n} (x_{jk} - z_{ik})^2 + \gamma_i \sum_{p=k+1}^{m} \delta(x_{jk}, z_{ik})}$$



Initialize the initial k prototypes, $Z_1, Z_2, \ldots, Z_n$, as cluster centers in each cluster.

Calculate the initial distance to each cluster for each observation to determine the similarity for each observation

Group each observation into each k cluster based on the closest distance to the initial initial.

Repeat steps 1 to 3 until there is no movement of an observation to another cluster

Mix Type DBSCAN

1. Gower's Distance

Gower distance is used to calculate the distance of mixed-sized data, namely numerical and categorical

$$\delta_G(X,Y) = \frac{\sum_{j=1}^{m} w_j f_j(x_j, y_j)}{\sum_{i=1}^{m} w_j}$$

SIMPLY EXAMPLE

ID	Tinggi (cm)	Warna Bunga	Berat (kg)
1	150	Merah	0.5
2	160	Kuning	0.6
3	170	Merah	0.55

Step 1, normalized numerical variable

- $z_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}$

Step 2, transformation categorical variable

- Give 0 value if the category is equal, 1 if different.
- For example: Merah vs Merah = 0, Merah vs Kuning = 1

Step 3, count the distance each individu

$$d_{ij} = \frac{1}{p} \sum_{k=1}^{p} s_{ijk}$$

For Example the count of individu 1 vs individu 2

Variabel Tinggi: $z_{11} = 0$, $z_{21} = \frac{10}{20} = 0.5$

Variabel Warna Bunga: Merah vs Kuning = 1

Variabel Berat: $z_{13} = 0$, $z_{23} = 1$

Gower Distance: $d_{12} = \frac{1}{3}(0.5 + 1 + 1) = 0.833$

Step 4, Store to Matrix

	1	2	3
1	0	0.833	0.75
2	0.833	0	0.417
3	0.75	0.417	0

Mix Type DBSCAN

2. DBSCAN *Clustering*

- Density-based spatial clustering of applications (DBSCAN) is a cluster algorithm that uses density to identify clusters based on the epsilon parameter and minimum samples in one cluster.
- Core is an object that meets the desired epsilon and minimum samples in one cluster.
- Border is an object adjacent to the core
- Noise is an object that is not included in the cluster and remains at the end of grouping

Set epsilon with KNN and set minimum points

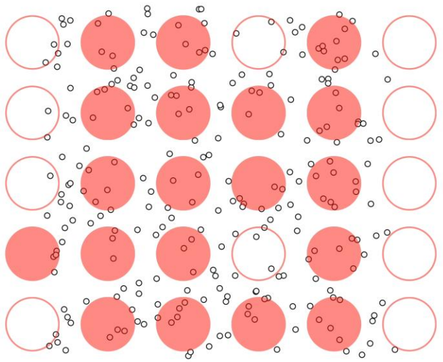
Set initials value

Calculate the Gower distance matrix which is used to calculate the distance in determining members of a cluster.

Cluster expansion where this step searches for all points within the epsilon radius.

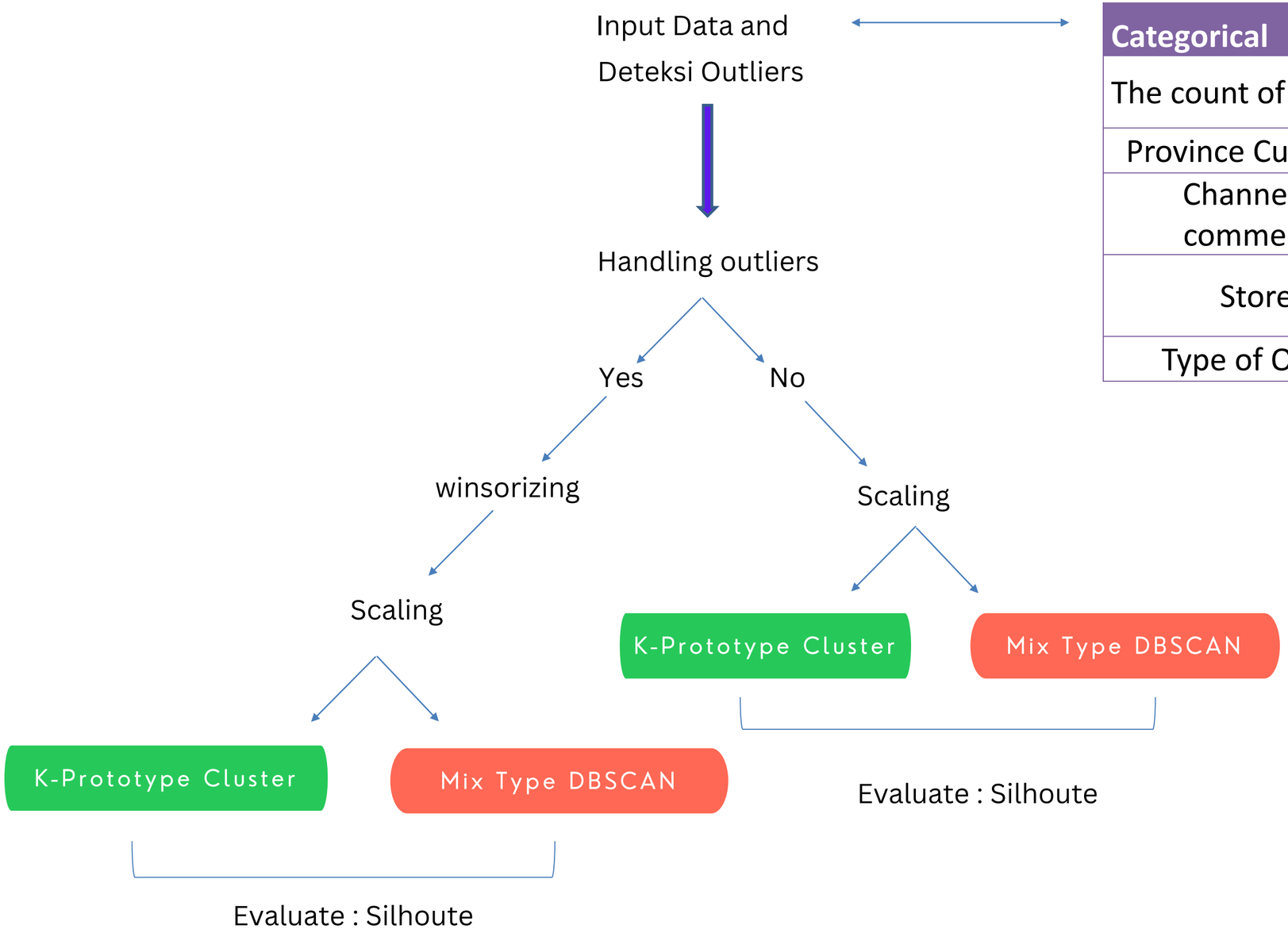
Handling noise or what is also commonly called outliers can be handled using the winsorizing method.

	1	2	3
1	0	0.833	0.75
2	0.833	0	0.417
3	0.75	0.417	0



Method

Flow Chart Study

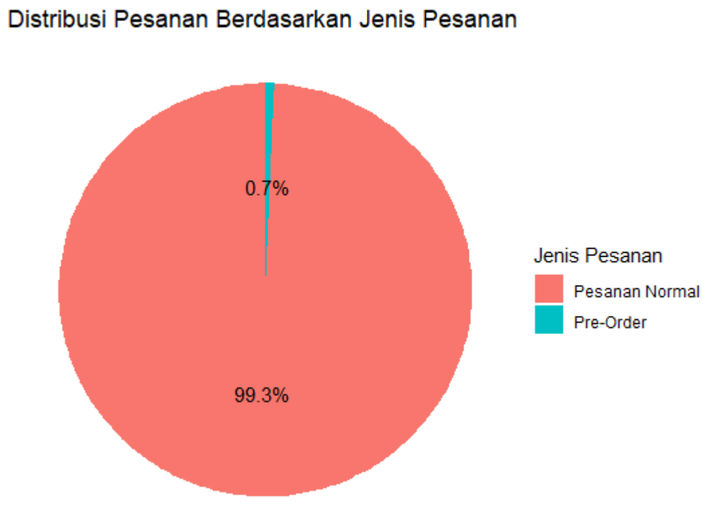


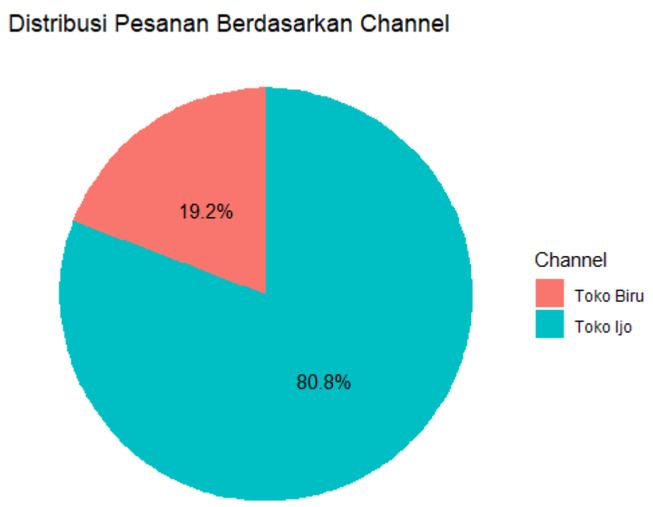
Categorical	Numerical
The count of product	Total Price (Rp)
Province Customer	Total mass (g)
Channel E-commerce	Voucher Marketplace
Store	Discount Cost of delivery
Type of Order	Discount from Store

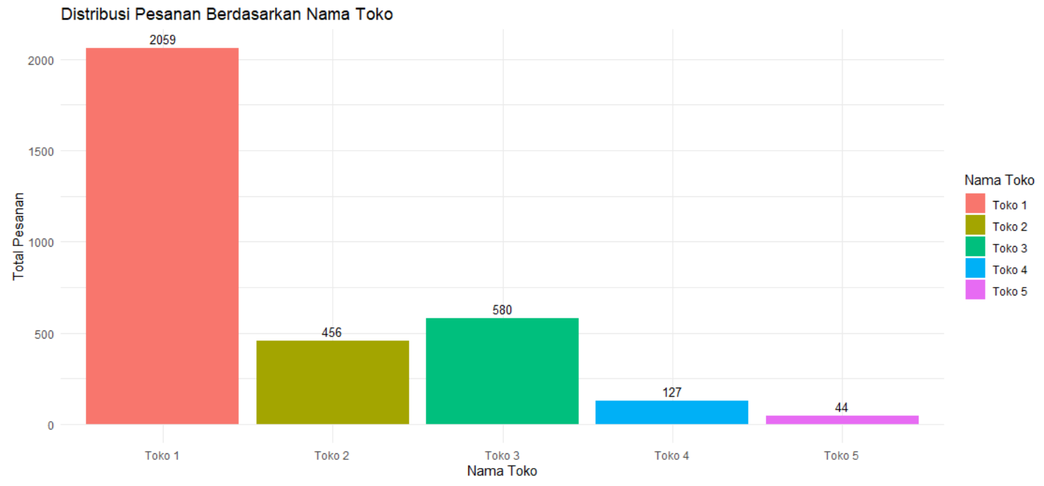
Online sales data from Company X starting from March 1, 2024, to May 15, 2024, where the total data is 3266 samples.

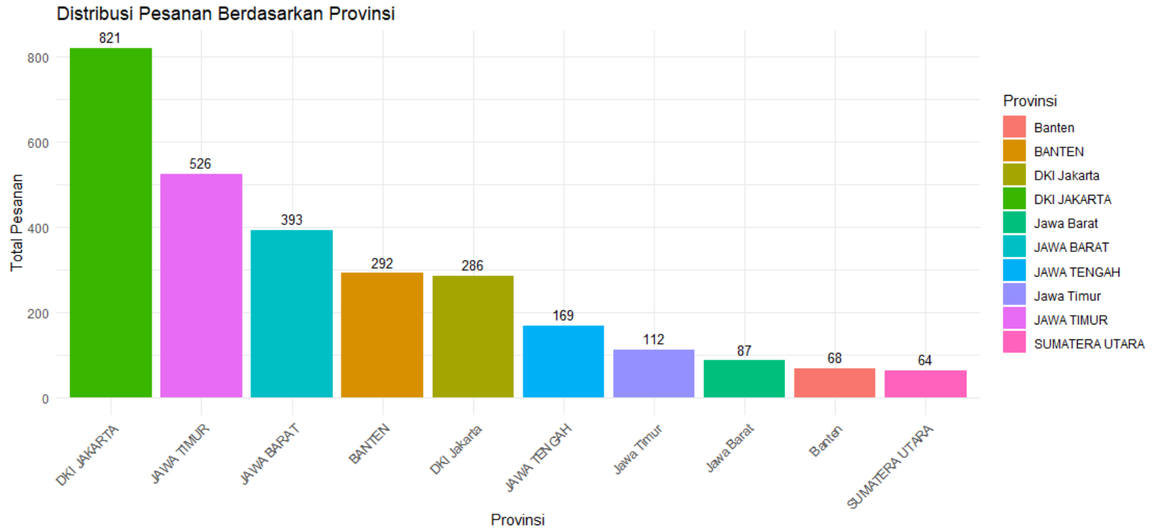
Result

Descriptive Data



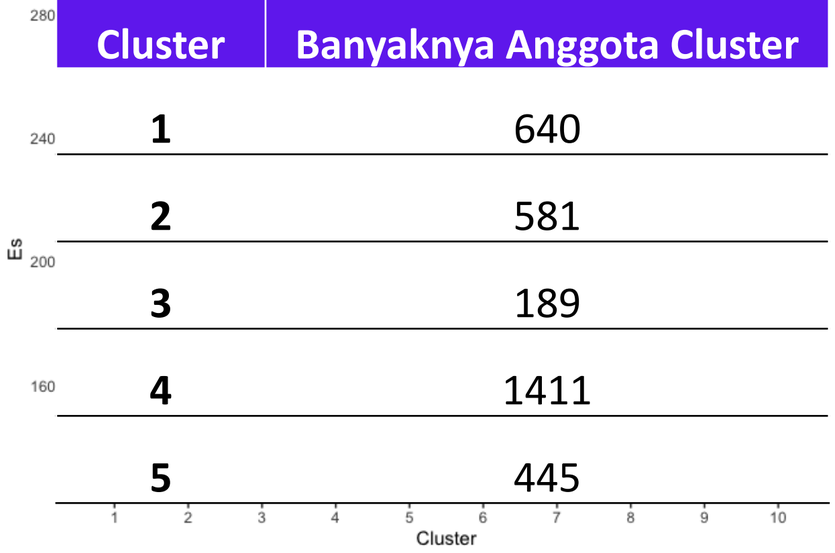






K-protoype

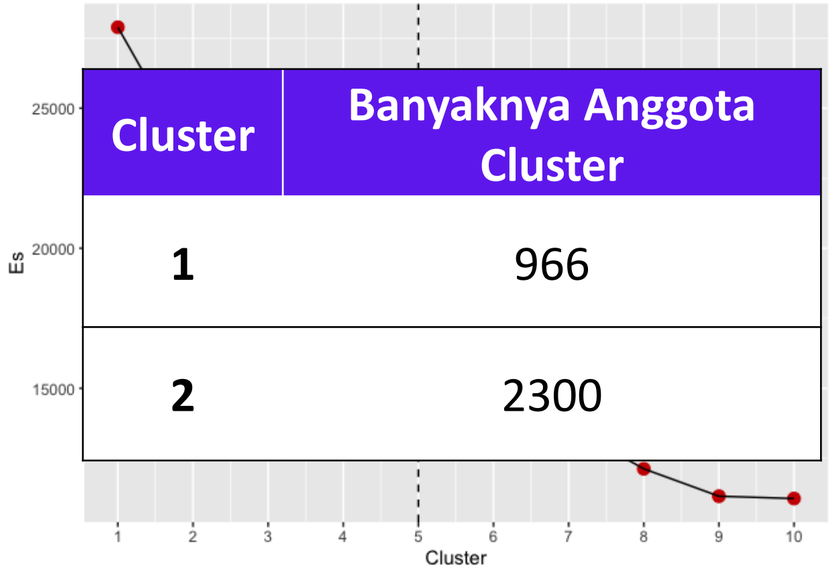
WITHOUT OUTLIERS



Cluster	Banyaknya Anggota Cluster
1	640
2	581
3	189
4	1411
5	445

Based on the Elbow chart, the optimal number of clusters is 2.

WITH OUTLIERS

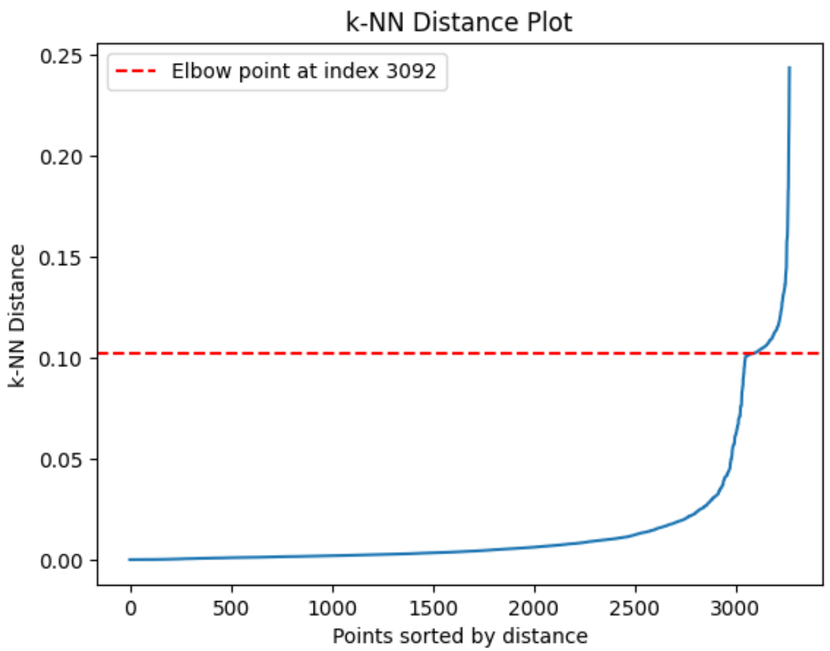


Cluster	Banyaknya Anggota Cluster
1	966
2	2300

Based on the Elbow chart, the optimal number of clusters is 5.

DBSCAN

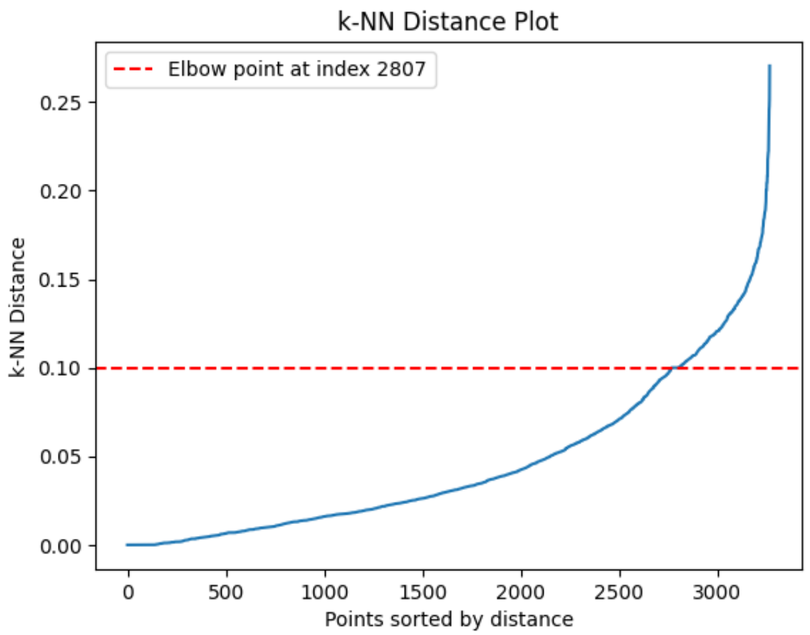
WITHOUT OUTLIERS



Epsilon = 0,10 minimum sample =10

Cluster	Banyaknya Anggota Cluster
1	2545
2	527
3	294

WITH OUTLIERS



Epsilon = 0,10 minimum sample =10

Cluster ke	Banyaknya Anggota Cluster
1	385
2	493
3	2388

Evaluation

Silhouette Coefficient measures the between-cluster distance against within-cluster distance. A higher score signifies better-defined clusters.

This measure has a range of [-1, 1]. Silhouette coefficients close to + 1 imply that the sample is far from neighboring clusters. A value of 0 means that the sample is on or very close to the decision boundary between two neighboring clusters, and negative values suggest that the samples may have been allocated to the wrong cluster

Outliers	K-Prototype	MT-DBSCAN
Ya	0, 21	0,52
Tidak	0, 41	0,33

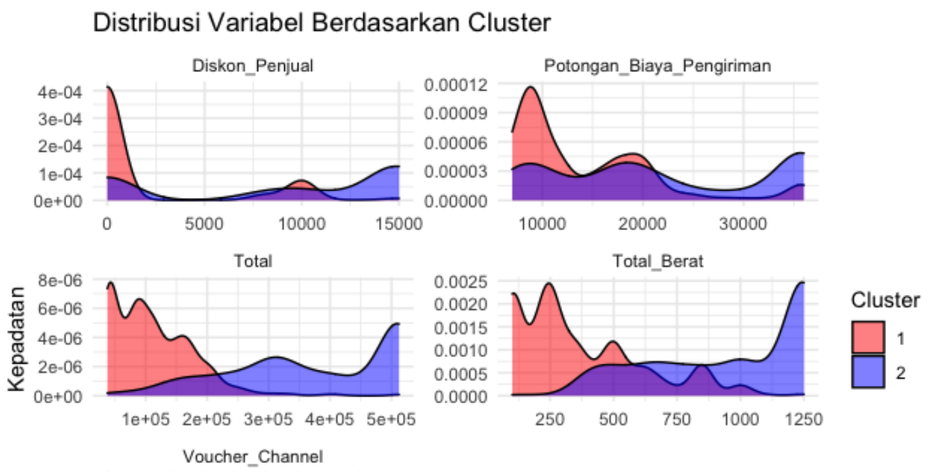
These results suggest that:

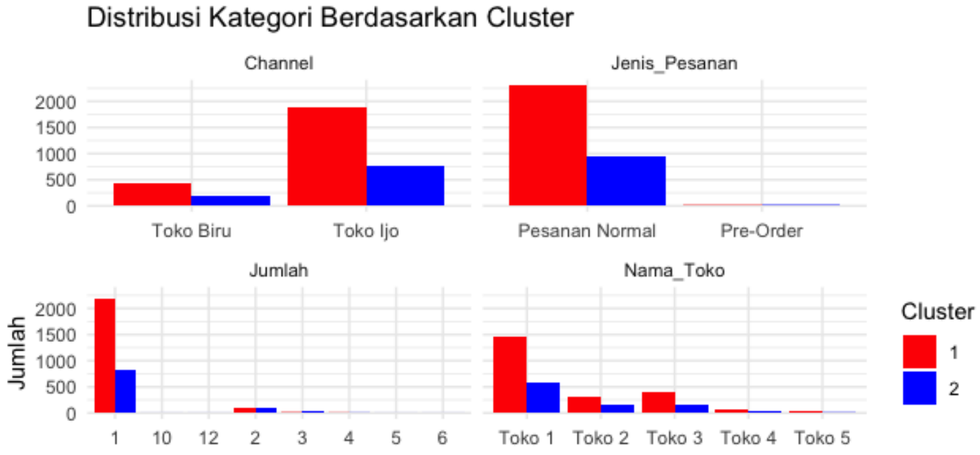
•DBSCAN performs better with outliers because it can identify and exclude them from the clusters.

•K-Prototypes performs better without outliers, as it relies on centroids that can be heavily influenced by outliers.

Visualization for Best Method

With Handling Outliers : K-Prototype



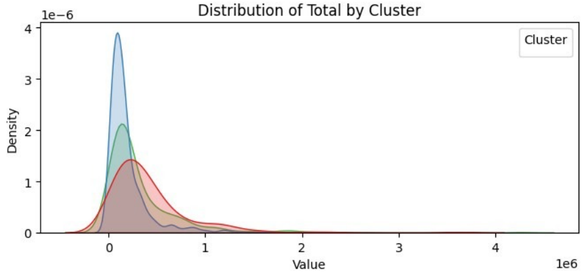


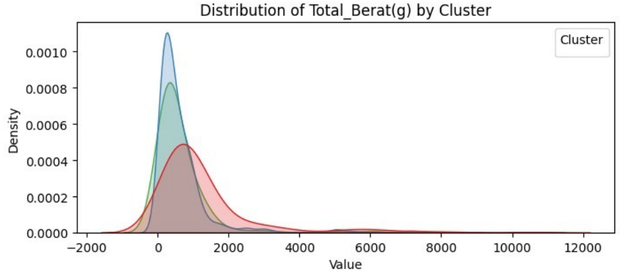
General Customer Behavior

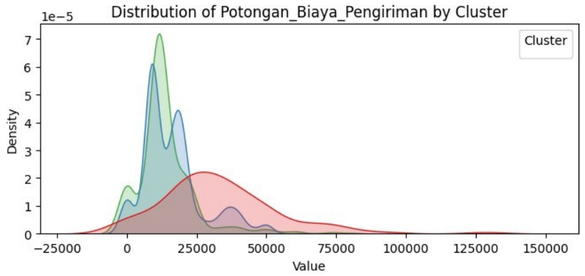
- **Cluster Characteristics**:
- **Cluster 1**: Represented by red in charts.
  - Lower nominal total transactions.
  - High purchase frequency.
  - Represents "Loyal Customers" or "Frequent Visitors".
- **Cluster 2**: Represented by blue in charts.
  - Fewer transactions compared to Cluster 1.
  - Higher transaction value.
  - Represents "Big Spenders".

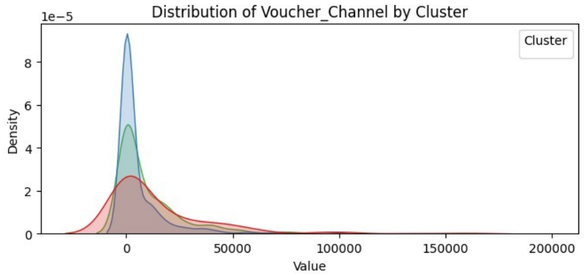
Visualization

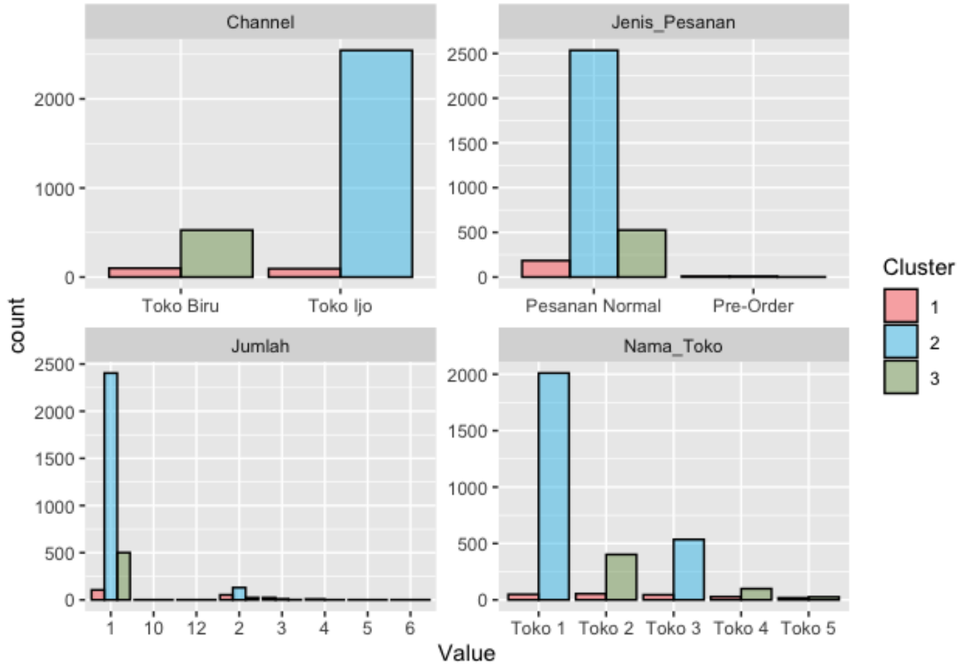
Without Handling Outliers : Gower Distance DBSCAN

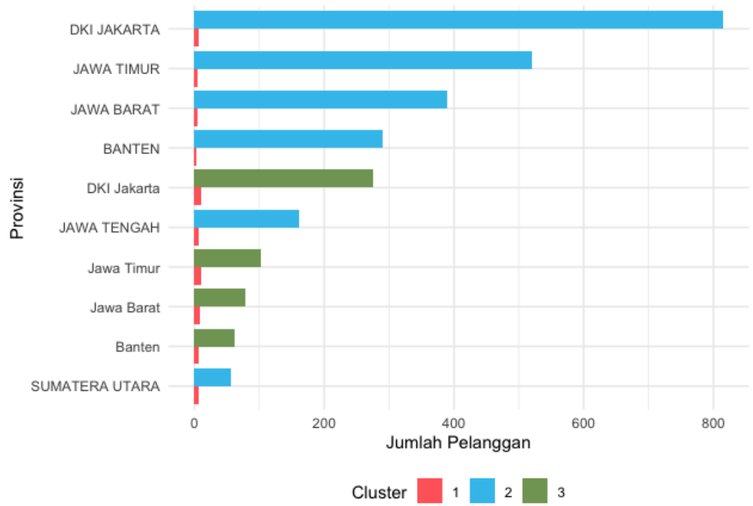












Summary

- Data containing outliers can be grouped using the MT-DBSCAN method, resulting 3 clusters. The MT-DBSCAN method is more effective when the data contains outliers because it utilizes Gower's Distance, which is robust against outliers.
- If the outliers in the data are addressed or the data does not contain outliers, the better clustering method is K-Prototype, which results 2 clusters.

1ST CLUSTER

2ND CLUSTER

3RD CLUSTER

Customers with high loyalty levels and frequent visits to the e-commerce platform channel (Loyal Customers). Policy recommendations for Company X are:

- Providing exclusive discount offers for loyal customers.
- Creating loyalty programs that provide incentives for customers to continue shopping (e.g., giving points for every transaction).
- Giving rewards to loyal customers in the form of special recognitions, such as discount vouchers for specific events like double dates or major holidays.
- Offering premium membership with additional benefits, such as a priority service team to handle transaction issues.

1ST CLUSTER

2ND CLUSTER

3RD CLUSTER

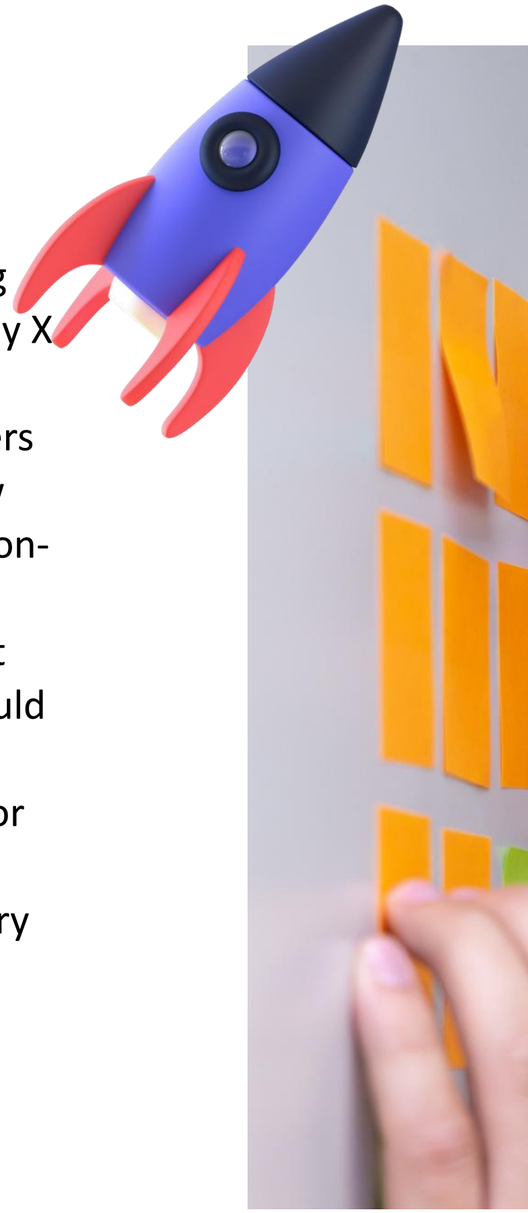
Customers with higher loyalty levels than Cluster 1 but not as high as Cluster 3. Policy recommendations for Company X related to customers in Cluster 2 are:

- Creating a premium membership program that provides special offers not available to lower-tier customers.
- Offering premium customer service such as a dedicated support team, special hotlines, or priority issue handling services.
- Actively engaging with customers in this cluster through surveys, feedback, or direct discussions to gather input and suggestions

1ST CLUSTER

2ND CLUSTER

3RD CLUSTER



Customers with very high purchase volumes (Big Spenders). Policy recommendations for Company X related to customers in Cluster 3 are:

- Creating a VIP membership program that offers exclusive benefits such as early access to new products and special offers not available to non-VIP customers.
- Offering a personal concierge service to assist with specific customer needs. This service could include purchase assistance, product recommendations, and handling complaints or special requests.
- Providing premium gifts, which could be luxury products or high-value shopping vouchers.

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Thank you